17 th INTERNATIONAL BIOLOGY OLYMPIAD
9-16 JULY 2006
Río Cuarto – República Argentina

PRACTICAL TEST

1

Plant Anatomy, Systematics and Physiology

Student code:	

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General remarks about the practical tests

DEAR PRATICIPANTS

The practical tests are organized in four different laboratories.

Nº 1- Plant Anatomy, Systematics and Physiology

Nº 2- Animal Anatomy, Systematics and Ecology

Nº 3- Biochemistry

Nº 4- Microbiology

- You have **1 hour** in laboratory Nº 1
- You have **1 hour** in laboratory Nº 2.
- You have **1 hour 30 minutes** in laboratory Nº 3,
- and **1 hour 30 minutes** in laboratory Nº 4.
- You can score maximum **40 points** in each laboratory, which means a total of **160 points** for the practical test.

Good luck !!!!!!!

Practical test Nº 1: Plant Anatomy, Systematics and Physiology

In this laboratory task you will have to work on the morphological, taxonomic and physiological aspects of higher plants in an integrated way.

Aims

A) To identify and compare vegetative organs.
B) To identify different taxa.
C) To relate leaf anatomy to photosynthetic pathways.

Materials:

- 5 samples (labeled 1-5).
- 5 slides.
- 5 coverslides.
- 1 razor blade.
- 1 felt-tip marker for glass.
- 1 tweezers.
- 2 histological needles.
- 1 dropper with distilled water and glycerin.
- 1 Petri dish with Safranin solution (it stains lignin).
- 1 Petri dish with distilled water.
- 1 microscope.
- Figure 2: microphotographs with details of leaf sectors.

Procedure

- Cut cross sections of sample 1.
- Place the sections into the Safranin solution.
- Transfer the sections to the Petri dish with distilled water to remove the excess of stain.
- Place the sections on a slide with water and glycerin and cover with a coverslide.

Repeat the procedure to obtain histological slides of the remaining samples.
Observe the obtained histological slides with the microscope. Remember to start observing with the lowest magnification power and then, end up with the 40x objective lens.

After examining each specimen and your prepared histological slide answer the following questions:

Q1 : Fill in the organ code number in the appropriate box.

Codes:

01- stem.
02- root.
03- leaf.
04- rhyzome

Sample	1	2	3	4	5
Code					

Q2 : Identify the taxon to which each sample belongs, to write its number in the appropriate box.

Taxon	Sample number:
Ginkgophyta	
Pinophyta	
Cycadophyta	
Magnoliophyta - Magnoliopsida	
Magnoliophyta-Liliopsida	

Q3 : The endodermis is a layer of cells that performs an important physiological role. Indicate with an “X” the sample/s where this cellular layer is observed.

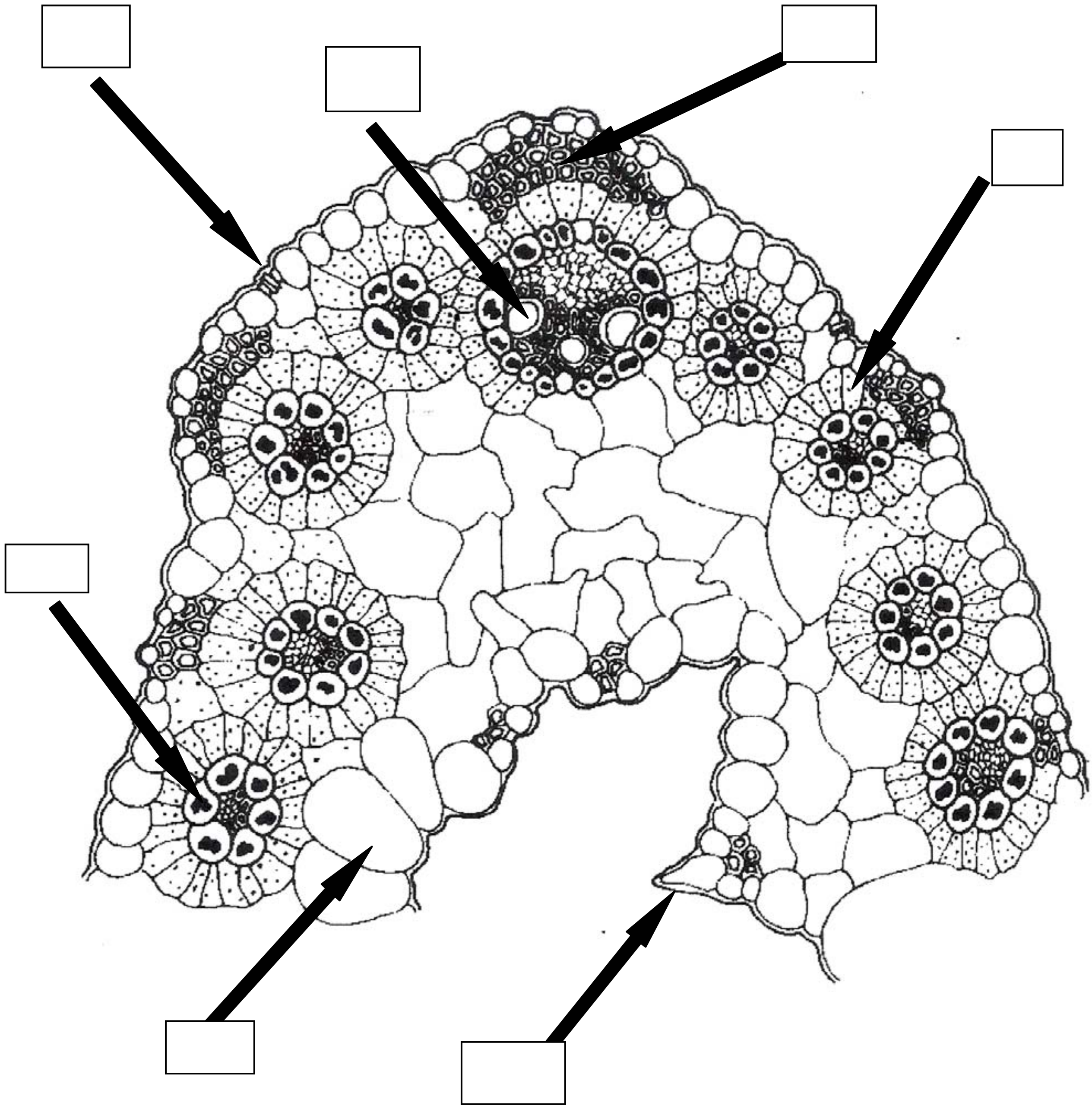
Sample	1	2	3	4	5

Q 4: Plants may differentiate collenchyma and sclerenchyma as supporting tissues. Both tissues show particular cytological characteristics that allow us to identify them. Circle the option that contain/s the sample number/s where collenchymatic tissue is observed.

a) 1, 2, 3.
b) 4, 5.
c) 4.
d) 2.
e) 1, 4.

Q 5: Examine carefully the leaf anatomical structure represented in the **figure Nº 1.**

Figure Nº 1: leaf anatomical structure



Could this leaf structure correspond to some of the organs previously cut and be part of the same plant? Circle the correct option.

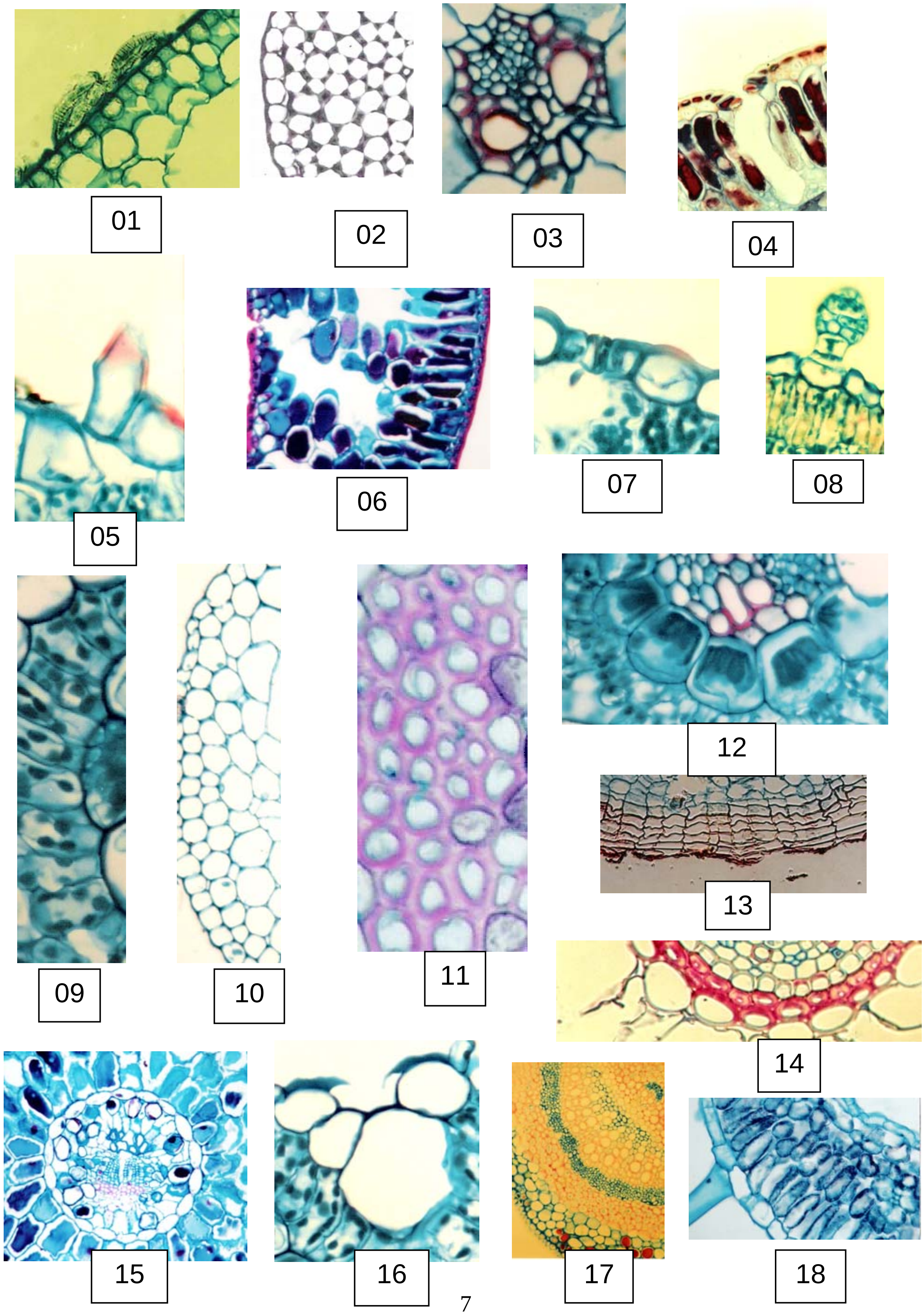
YES NO

If the answer was affirmative, indicate with an “X” the corresponding sample/s.

Sample	1	2	3	4	5

Q 6: Microphotographs with details of leaf sectors are shown **(Figure nº 2)**. Select the codes of those microphotographs corresponding to the sectors pointed out in the leaf diagram represented in **Figure nº 1**

Figure Nº 2: microphotographs with details of leaf



Q7: Keeping in mind all the anatomical characters observed when completing **figure 1** you could infer that this leaf corresponds to a species that belongs to the Family (Circle the correct answer):

a) Liliaceae.
b) Fagaceae.
c) Brassicaceae.
d) Poaceae.
e) Araceae.

Q8: Leaf anatomy is related to the environment where the plant grows and it indicates its photosynthetic pathway. Keeping this in mind, observe again the leaf structure represented in **figure 1** and select the codes that correspond to this structure.

01- It follows the Calvin Cycle photosynthetic pathway or C3 pathway.
02- It has an additional method of fixation of the carbon (not alternative) that works separately from the Calvin Cycle.
03- It shows a stratified mesophyll.
04- It shows a radiated mesophyll (Kranz).
05- It shows chloroplast dimorphisms and /or sizes.
06- Optimum temperature for photosynthesis is between 15-25º.
07- Optimum temperature for photosynthesis is between 30º-45º.
08- It shows two well-developed sheaths around the vascular bundles.
09- It shows one sheath around the vascular bundles.
10- It does not show sheaths around the vascular bundles.
11- The decarboxylation phase takes place in different structures of the leaf.

Answer: ....

Q9: Complete the following comparative table of the three main photosynthetic pathways of carbon assimilation, keeping in mind the codes for each character.

Enzyme responsible for the initial carboxylation:

01- Ribulose 1,5 - bisphosphate carboxylase-oxygenase (Rubisco).
02- Phosphoenolpyruvate carboxylase (PEPase).
03- Sucrose-phosphate synthase (SPase).
04- Rubisco and PEPase.
05- SPase and PEPase.

Leaf anatomy:

01- stratified.
02- Kranz structure (radiated).
03- succulent.

The CO_2 fixation time:

01- Day.

02- Night.
03- Day and night.

First stable product of $C0_2$ fixation:

01- Made up of six carbons.
02- Made up of four carbons.
03- Made up of three carbons.

Efficiency in water use:

01- Medium.
02- High.
03- Low.

Photosynthetic rate:

01- Medium.
02- High.
03- Low.

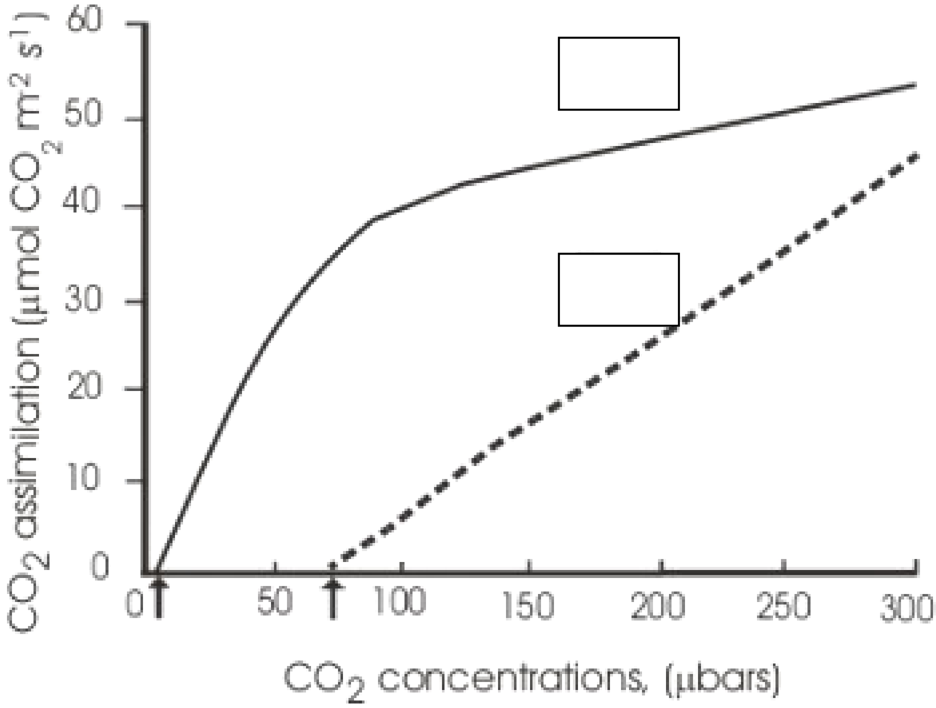
Character	C3	C4	CAM
Enzyme responsible for the initial carboxylation:			
Leaf anatomy:			
The CO_2 fixation time:			
First stable product of CO_2 fixation:			
Efficiency in water use:			
Photosynthetic rate:			

Q10: If a plant is placed into a closed chamber and exposed to the light, it is observed that the CO_2 concentration in the air inside the chamber decreases for a while due to photosynthesis. The decrease is gradual but it never reaches the zero value. A balance is reached between the CO_2 captured by photosynthesis and the one released by respiration and photorespiration. This balance is known as CO_2 compensatory point.

In the following graph the effect of the atmospheric concentration of CO_2 on the photosynthesis rate in plants C3 and C4 is observed. The arrows indicate the compensatory points of each plant.
Indicate the curve that corresponds to each plant by writing C3 or C4 in the right box.

Reference:

↑: **compensatory points of each plant**



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PRACTICAL TEST

2

Animal Anatomy, Systematics and Ecology

Student Code:	

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General remarks about the practical tests

DEAR PARTICIPANTS

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Nº 2- Animal Anatomy, Systematics and Ecology

Nº 3- Biocheminstry

Nº 4- Microbiology

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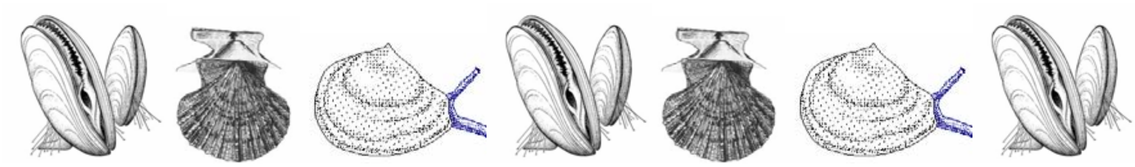
Good luck !!!!!!!



Practical Test Nº 2: Animal Anatomy, Systematics and Ecology

Introduction

Bivalves are an important group of molluscs, the second in number of species after gastropoda. Other names for the class include Pelecypoda, and Lamellibranchia.

Bivalves include all laterally compressed mollusc species; they typically have two-part shells dorsally hinged by strong muscles and ligaments.

The mantle, which secretes the shell, is the dorsal body wall covering the visceral mass. The mantle cavity is lateral and in most bivalves the gills have a respiratory and digestive function.

Unlike other molluscs, bivalves lack a radula and feature labial palps which carry food from the gills to the mouth.

The head is small and it does not feature specific sensory organs.

Task Nº 1: Bivalve dissection (13 points)

Task Nº 1 includes part A (10 points) and part B (3 points)

<u>The aim</u> is to compare anatomical structures in three marine bivalves.

<u>Materials:</u>

- ✓ Tray containing three samples of marine bivalves numbered 1, 2 and 3 (stored in 70% alcohol).
- ✓ 1 dissection table.
- ✓ 1 lancet.
- ✓ 1 tweezers.
- ✓ 10 color pins (3 green, 3 red, 3 blue and 1 yellow).
- ✓ 1 pair of disposable gloves.
- ✓ 1 respirator mask.
- ✓ 1 magnifying glass.

REMARK: BEFORE STARTING THE PRACTICAL TASK, BE SURE TO HAVE ALL THE LISTED MATERIALS, OTHERWISE RAISE YOUR HAND TO CALL THE ASSISTANT.

PART A

Procedure

1- Put on the gloves and respiratory mask.
2- Before starting the dissection, locate the external parts of the bivalve (Figure 1).

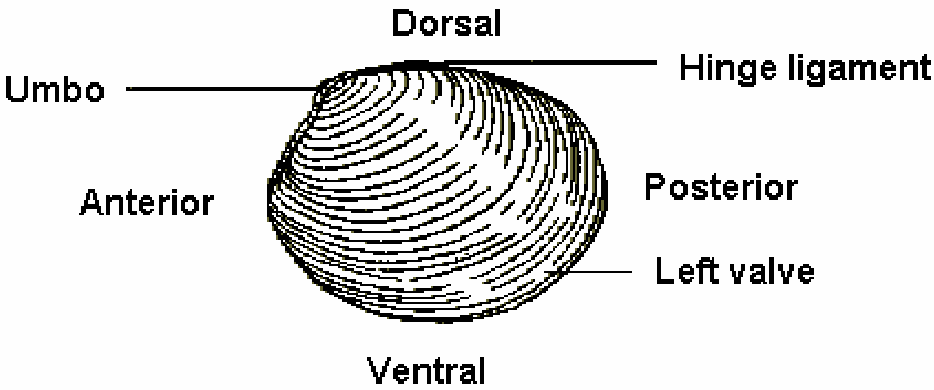


Figure 1

3- The valves are hinged by ligaments. In order to identify the internal structure you have to dissect the bivalve. You must be **very careful** when separating the valves so as not to hurt your hands.

Insert the lancet (Figure 2) and cut ✂, the adductor muscle/s, according to the bivalve.

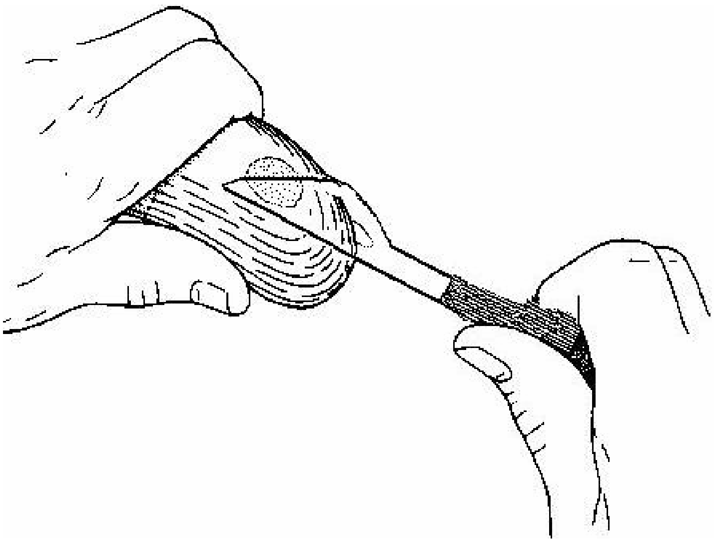


Figure 2

4- In order to separate the valves completely, once the muscle/s is/are cut, you must cut carefully the ligament in the umbo area.

5- Once the three samples are dissected, identify the structures with different color pins.

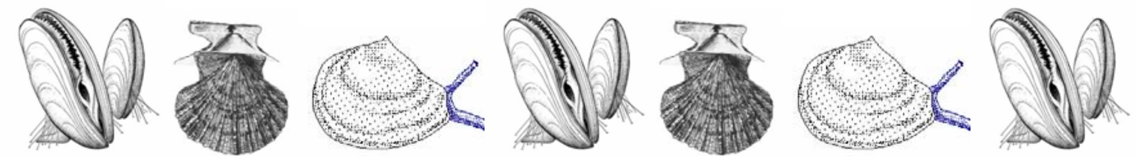
Use 3 pins for each bivalve sample (green, red, and blue), and a yellow pin only for bivalve 2, in the following way:

- **green** pin for the **foot**.
- **red** pin for **labial palps**.
- **blue** pin for the **gills**.
- **yellow** pin for the **inhaling siphon**. (<u>Only for bivalve 2)</u>

6- **After finishing the task you must raise your hand. An assistant will check the task. The Practical Test Sheet should be signed by both, you and the assistant**.

Signatures:

Student.:...................................**Assistant**:...



PART B

As you have seen during the dissection, the three bivalves show differences in their muscles.

There exists a muscle classification according to their number and size:

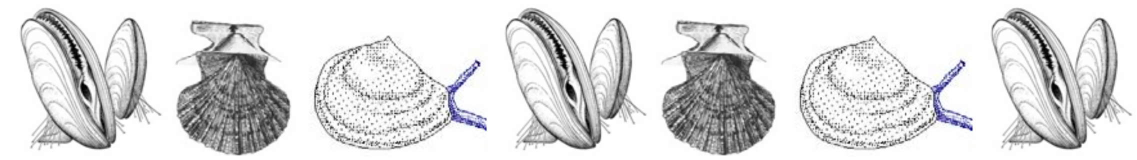
- ✓ Dimyarian isomyarian condition: in which both muscles have similar size.
- ✓ Dimyarian heteromyarian condition: in which both muscles are different in size.
- ✓ Monomyarian condition: Having only one, large adductor muscle to close the valves.

Complete the table by using the codes below.

	Bivalve 1	Bivalve 2	Bivalve 3
Condition			

Codes:

01- Dimyarian isomyarian.
02- Dimyarian heteromyarian.
03- Monomyarian.



Task Nº 2: Bivalve adaptive radiation

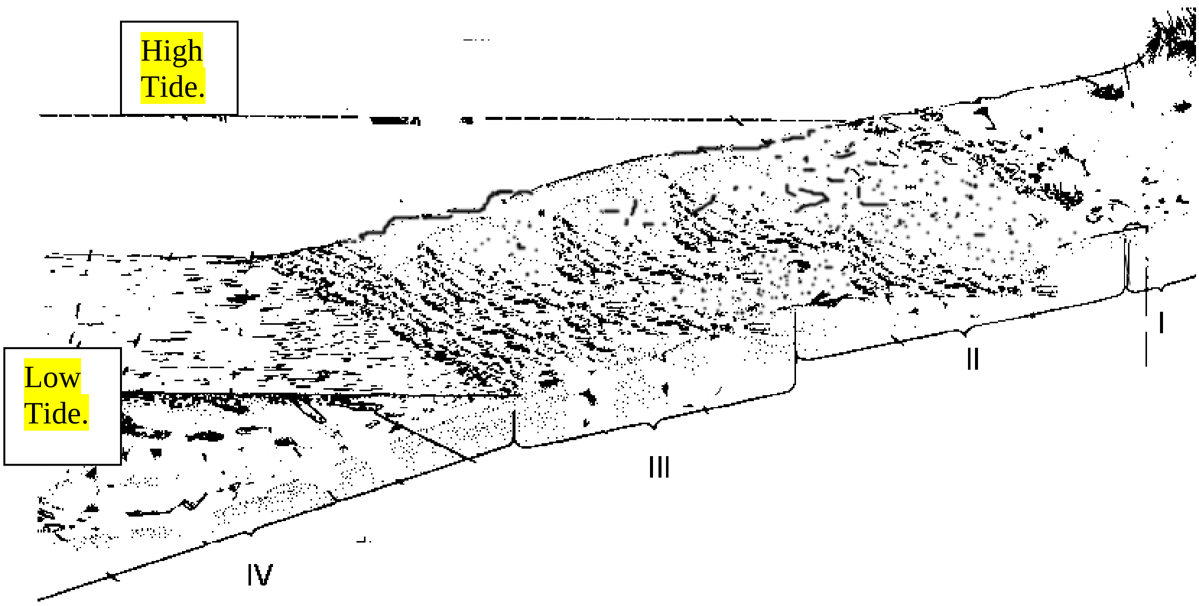
Task Nº 2 includes parts A, B and C.

Most bivalves are filter feeding molluscs, that is to say, they filter marine water to obtain their food consisting mainly of plankton and suspended organic matter. The evolutionary acquisition of feeding by filtering allowed them to colonize many habitats, thereby giving rise to an important adaptative radiation.

The aims of this part of the test are to determine the habitat of the marine bivalve samples and to identify the exomorphological and anatomophysiological characteristics associated to these habitats.

PART A (9 points) – Below there are two marine zones, one corresponding to a sandy beach (Figure 1), and the other to a rocky beach (Figure 2).

Figure 1



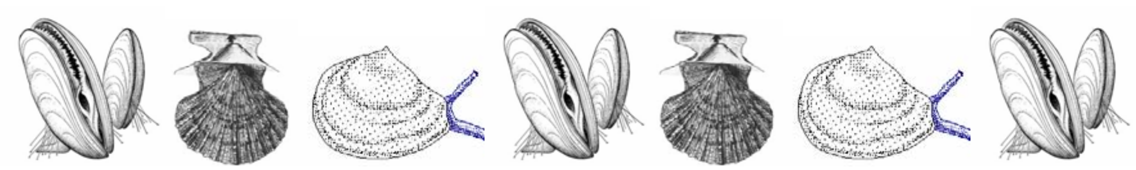
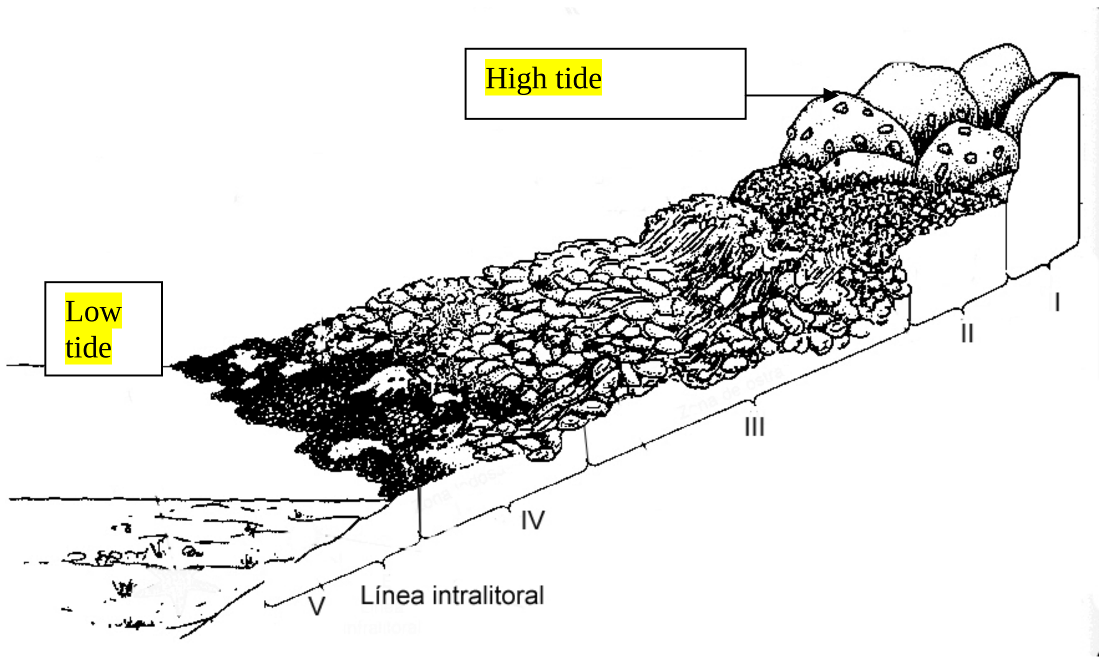


Figure 2

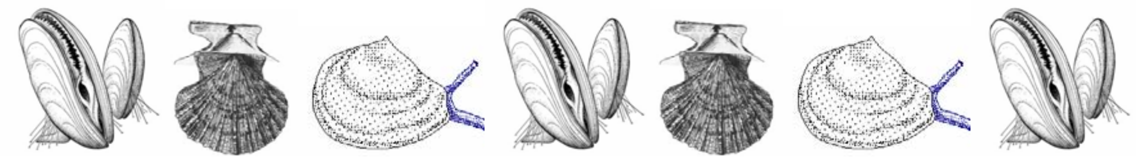


Fill in the corresponding box in each table, indicating the site where the samples given in this practical task can be found.

Codes:

01- Bivalve 1. 02- Bivalve 2. 03- Bivalve 3.

Sandy beach	Zone I	Zone II	Zone III	Zone IV



Rocky beach	Zone I	Zone II	Zone III	Zone IV	Zone V

PART B (6 points)- Keeping in mind the zones occupied by bivalves in rocky and sandy beaches, you must determine the category of the given samples by writing an **"X"** in the corresponding box.

	Bivalve 1	Bivalve 2	Bivalve 3
Burrowers in soft substrate INFAUNA			
Surface dwellers attached to the substrate EPIFAUNA			
Free swimming			

PART C (12 points)- A series of characteristics related to the three given bivalves and their habitats is given below. Examine your dissected specimens and summarize <u>ALL</u> their characteristics by writing the appropriate answer codes from the list into the table below.

Answer code:

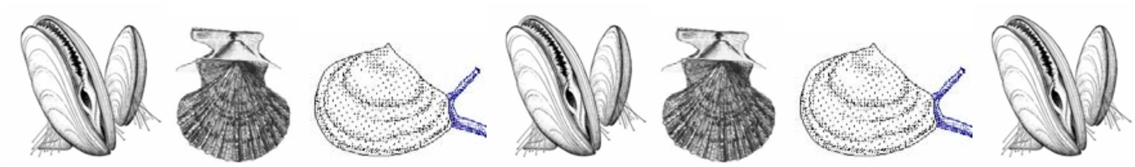
01- large, burrowing foot

02- reduced, finger-like foot.

03- highly reduced and barely visible foot.

04- no anterior adductor muscle.

05- no siphons.



06- two siphons: incurrent and excurrent (inhaling and exhaling).

07- fringed incurrent siphon.

08- highly developed sensory lobes in the mantle, with tentacles and small ocella.

09- flat lower valve (right)

10- mantle edge with fusion points.

11- byssal threads.

Bivalve 1	Bivalve 2	Bivalve 3

Practical Test Nº 2: Animal Anatomy, Systematics and Ecology

TASK Nº 1: Bivalve dissection (13 points)

PART A: One point per pin. Total: 10 points.

PART B: One point per box. Total: 3 points

	Bivalve 1	Bivalve 2	Bivalve 3
Condition	**02**	**01**	**03**

Task Nº 2: Bivalve adaptive radiation (27 points)

Part A: One point per box. Total: 9 points

Sandy beach	Zone I	Zone II	Zone III	Zone IV
	02	**02**	**02**	**03**

Rocky beach	Zone I	Zone II	Zone III	Zone IV	Zone V
	01	**01**	**01**	**01**	**03**

Part B: Two point per each correct mark .Total: 6 points.

	Bivalve 1	Bivalve 2	Bivalve 3
Burrowers in soft substrate INFAUNA		**X**	
Surface dwellers attached to the substrate EPIFAUNA	**X**		
Free swimming			**X**

Part C: One point per correct code. Total: 12 points

Bivalve 1	Bivalve 2	Bivalve 3
02- 05- 10- 11	**01- 06- 07**	**03- 04- 05- 08- 09**

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PRACTICAL TEST

3

Biochemistry

Student code:	

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General remarks about the practical tests

DEAR PARTICIPANTS

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Nº 1- Plant Anatomy, Systematics and Physiology

Nº 2- Animal Anatomy, Systematics and Ecology

Nº 3- Biochemistry

Nº 4- Microbiology

- You have **1 hour** for each laboratory: Nº 1 and Nº 2.
- You have **1 hour 30 minutes** for each laboratory: Nº 3 and Nº 4.
- You can score maximum of **40 points** in each laboratory, which means a total of **160 points** for the whole practical test.

Good luck !!!!!!!

Practical test Nº 3: Biochemistry

Enzymatic determination of glucose

TASK 1: You have to perform a calibration curve using a standard of glucose, with known concentration. Then, plot the results as absorbance versus glucose concentration (15 points)

Important: Raise the red card to call the lab assistant when you are ready to use the spectrophotometer.

Introduction:

Glucose oxidase (GOD) catalyzes the oxidation of (beta)-D-glucose to D-gluconic acid and hydrogen peroxide. It is highly specific for (beta)-D-glucose and does not act on (alpha)-D-glucose. The horseradish peroxidase (POD) breaks down hydrogen peroxide into water and oxygen, using the dye (4-aminophenazone) as an electron donor. At the same time, the dye is converted to its oxidized form, which is a colored compound. Since the amount of hydrogen peroxide produced indicates how much reaction has taken place, the formation of the red color can be used to follow the course of the reaction.

Its major use is in the determination of free glucose in body fluids. Although specific for (beta)-D-glucose, glucose oxidase can be used to measure total glucose, because as a result of the consumption of (beta)-glucose, (alpha)-glucose from the equilibrium is converted to the (beta)-form by mutarotation.

PRINCIPLE

The reaction system is as follows:

$$\text{Glucose} + O_2 + H_2O \xrightarrow{\text{GOD}} \text{Gluconic Acid} + H_2O_2$$

$$2\ H_2O_2 + \text{4-AP} + \text{4-Hydroxybenzoate} \xrightarrow{\text{POD}} \text{red quinoneimine}$$

Glucose oxidase reagent: solution containing glucose oxidase, peroxidase, 4-aminophenazone (4-AP), and phosphate buffer pH 7.0 containing hydroxybenzoate.

Reagents:

1. glucose oxidase reagent (ready to use).
2. glucose solution (unknown concentration).
3. glucose solution 5 mg. ml^{-1}.
4. distilled water.

Equipment

1. Lab gloves (1pair).
2. Marker pen (1).
3. 1.5 ml microtubes (18).
4. Pipettes (2).
5. Incubator at 37^{o}C .
6. Spectrophotometer (you will use it with lab assistants).
7. Spectrophotometric cuvettes (8).
8. Paper towels (3)
9. 1000 µl tips (30)
10. 200 µl tips (30)

Instruments:

Tip ejecting knob

Adjustable Wheel

Pushing Knob

Counter

Adjustment method

You have to pull up the Adjustable Wheel, then you can revolve the adjustable wheel or knob. Adjust the required volume and push down the Adjustable Wheel.

Remember that minimal and maximal volumes for P100 are 10 µl and 100 µl respectively.

For P1000 minimal volume is 100 µl and maximal volume is 1000 µl.

Usage method:

Please secure the suction tip, after that slightly push down the pushing knob to first stop, hold and immerse the tip into solution vertically. The immersed depth of the tip is 2-4 mm, then release the pushing knob slowly and make it return to the original position. Take off the pipette from the liquid and place the suction tip of the pipette into a special container receiving the dispensed liquid. The tip must be close to the inner wall of the container. Depress the pushing knob to the first stop and further more to discharge the solution completely from the tip. After that, you can take away the pipette and release the button. Eject the used tip to the trash recipient by pressing the Tip ejecting knob.

EXPERIMENTAL PROCEDURE

1) Label five 1.5-ml microtubes 1/2 through 1/32 with a marker pen. Using the glucose standard solution (5 mg. ml $^{-1}$) perform the following serial dilutions (in distilled water) in a final volume of 100 µl: 1/2, 1/4, 1/8, 1/16, and 1/32.

2) Mix well and perform (in a new 1.5-ml microtubes set) the enzymatic determination of glucose for each dilution according to the following squeme.

	1/2	1/4	1/8	1/16	1/32	Blank
Sample volume	10µl	10µl	10µl	10µl	10µl	0
Water volume	0	0	0	0	0	10 µl
Glucose oxidase reagent volume	1 ml	1 ml	1 ml	1 ml	1 ml	1 ml

3) Mix well and incubate microtubes at 37°C for 5 min.

4) Put the content of each microtube in a spectrophotometric cuvette.

5) Read absorbance in a spectrophotometer at 505 nm. You have to use the blank for calibration. (**When you are ready to read in the spectrophotometer, please call the lab assistant). If you have to wait to use the spectrophotometer, please continue with the questions. The extra waiting time will not affect the results.**

6) Plot the absorbance versus the amount of glucose (in µg) on the plotting paper below.

	Dilutions				
	1/2	1/4	1/8	1/16	1/32
glucose (µg in the reaction mix)					
Absorbance at 505 nm					

TASK 2: Determination of the glucose concentration in a sample employing the standard curve obtained before. (10 points)

1) Perform the glucose oxidase reaction to the glucose sample (unknown concentration), according to the follow scheme.

	Sample	Blank
Sample volume	10µl	0
Water volume	0	10 µl
Glucose oxidase reagent volume	1 ml	1 ml

2) Mix well and incubate microtubes at 37°C for 5 min.

3) Put the content of each microtube in a spectrophotometric cuvette

4) Read absorbance in the spectrophotometer at 505 nm. You have to use the blank for calibration. **(When you are ready to read in the spectrophotometer please call the lab assistant)**

5) Using the calibration curve, calculate the glucosa concentration of the sample in µg. ml^{-1}.

Absorbance of the sample	
Concentration of the sample (in µg . ml^{-1})	

~~**QUESTION 1:**~~ DELETED ~~As many glucose assays measure the peroxide produced by the glucose oxidase reaction, it is important that the enzyme used for these assays presents: **(1.5 points)**~~

~~A) a low catalase content.~~

~~B) a high catalase content.~~

~~C) a low peroxidase content.~~

~~D) a high peroxidase content.~~

~~WRITE DOWN THE LETTER CORRESPONDING TO CORRECT ANSWER~~

~~**Answer**:..~~

QUESTION 2: Glucose oxidase reagent may contain catalase. If such a condition is not taken into account the obtained results will give **(3 points)**:

A) underestimation of the glucose in the assay.

B) overestimation of the glucose in the assay.

C) no effect in the assay.

WRITE DOWN THE LETTER CORRESPONDING TO CORRECT ANSWER

Answer:..

QUESTION 3: The most favorable pH value (the point at which the enzyme is most active) is known as the optimum pH. Extremely high or low pH values usually result in a complete loss of enzyme activity due to **(1 point)**:

A) The breakdown of the secondary structure of the protein.

B) The breakdown of the tertiary structure of the protein.

C) The breakdown of the primary structure of the protein.

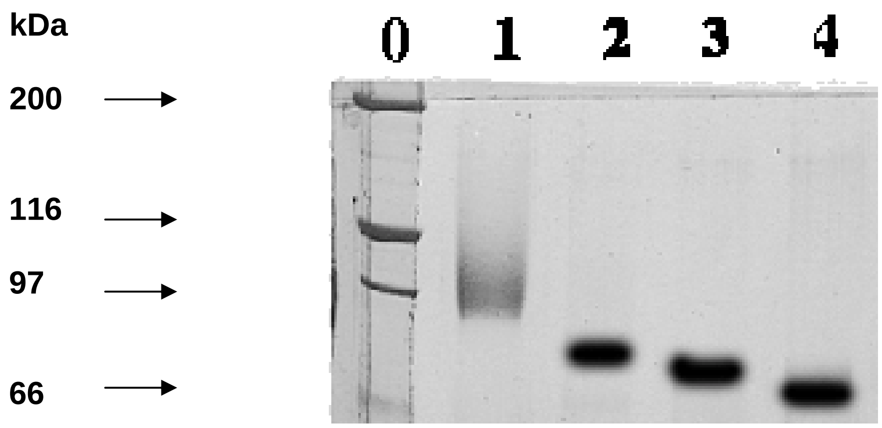
SELECT ONLY ONE CORRECT ANSWER. MARK IT WITH A CROSS.

☐ A. ☐ B. ☐ C. ☐ A, B

☐ A, C ☐ B, C. ☐ A, B, C.

QUESTION 4: Glucose oxidase from the fungus *Aspergillus niger* was overexpressed in yeast. The ~~recombinant~~ glucose oxidase was purified and glycosylation pattern was analyzed by treatment with endoglycosidase H and α-mannosidase. After treatment, an aliquot was used for SDS-PAGE (electrophoresis in polyacrylamide gels containing sodium dodecyl sulphate) in reducing conditions. The remaining enzyme was employed for determination of the K_M (Michaelis-Menten constant) with glucose as the substrate. Michaelis-Menten constant (K_M) is the concentration in moles/litre of a substrate at half the maximum velocity of an enzymatic reaction. **(7 points)**

The values of the *K*M for each glycoforms are shown below the figure 1.



*K*M (mM)	20	28	35	60

Figure 1: Analysis of the deglycosylation of Glucose Oxidase by 7.5% acrylamide SDS-PAGE gel electrophoresis. Lane 0 is the molecular mass standard. Lane 1 untreated enzyme. Lane 2 endoglycosidase H treated enzyme. Lane 3 α-mannosidase treated enzyme. Lane 4 endoglycosidase H and α-mannosidase treated enzyme (fully deglycosylated enzyme).

Which of the following is/are correct conclusion(s) from these results (figure 1)?

A) Glucose oxidase is a homodimer with a molecular mass of 96 kDa.

B) The deglycosylated form has a molecular mass of approximately 68 kDa.

C) Glucose oxidase is glycosylated since the treatment with endoglycosidase H and α-mannosidase results in a form with lower molecular mass.

D) The polysaccharide moiety of glucose oxidase contains *N*-acetylglucosamine and mannose.

MARK THE CORRECT ANSWER/ ANSWERS.

☐ A ☐ B ☐ C ☐ D

Which is/are correct conclusion(s) from the results obtained in the determination of KM for each glycoform?

A) The affinity of the fully glycosylated enzyme for glucose is higher than the affinity of the deglycosylated enzyme

B) The glucose oxidase activity is completely abolished in the deglycosylated form

C) The lack of the sugar moiety could cause changes in the structure of the active site of the enzyme resulting in the observed modifications of the KMs.

MARK THE CORRECT ANSWER/ ANSWERS.

☐ A ☐ B ☐ C

QUESTION 5: ~~Another treatment consisted in the purification of the recombinant glucose oxidase under non-denaturating conditions and in presence of glutaraldehyde~~. The purified enzyme was analyzed by SDS-PAGE in reducing (DTT+) as well as in non-reducing (DTT-) conditions ~~(without SDS)~~. ~~Prior to loading, the samples were resuspended in loading buffer with (+) and without (-) DTT (a reducing agent).~~ The obtained results are shown in figure 2 **(4 points)**

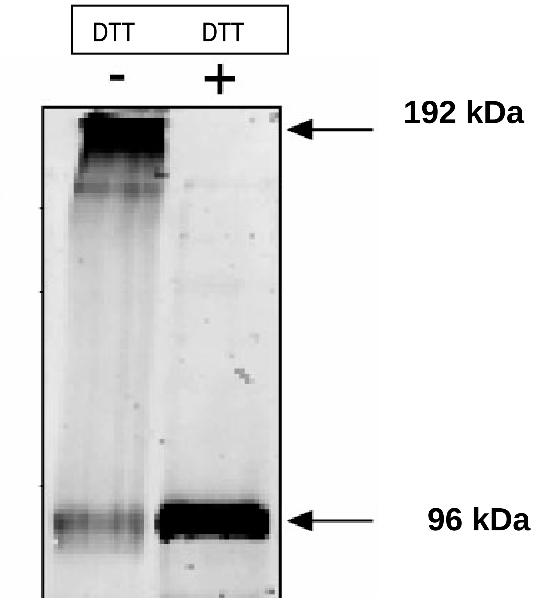


Figure 2: SDS-PAGE analysis of purified glucose oxidase. ~~(non-denaturating electrophoresis in polyacrylamide gels) of recombinant glucose oxidase purified under non-denaturating conditions.~~

Taking in account the results obtained from figure 1 and figure 2 the most probable conformation of the recombinant glucose oxidase is:

A) A monomeric enzyme non-glycosylated.

B) A monomeric enzyme glycosylated.

C) An homodimer consisting of two monomers both glycosylated.

D) An heterodimer consisting of two subunits one of them glycosylated.

MARK THE CORRECT ANSWER.

□ A □ B □ C □ D

17 th INTERNATIONAL BIOLOGY OLYMPIAD

Correct answer sheet
Practical test Nº 3: Biochemistry

Enzymatic determination of glucose

TASK 1: (15 points)

5 points	Dilutions				
	1/2	1/4	1/8	1/16	1/32
glucose (µg in the reaction mix) **0.5 pt each**	**25**	**12.5**	**6.25**	**3.125**	**1.56**
Absorbance at 505 nm **0,5 pt each**	**0.8**	**0.4**	**0.2**	**0.1**	**0.05**

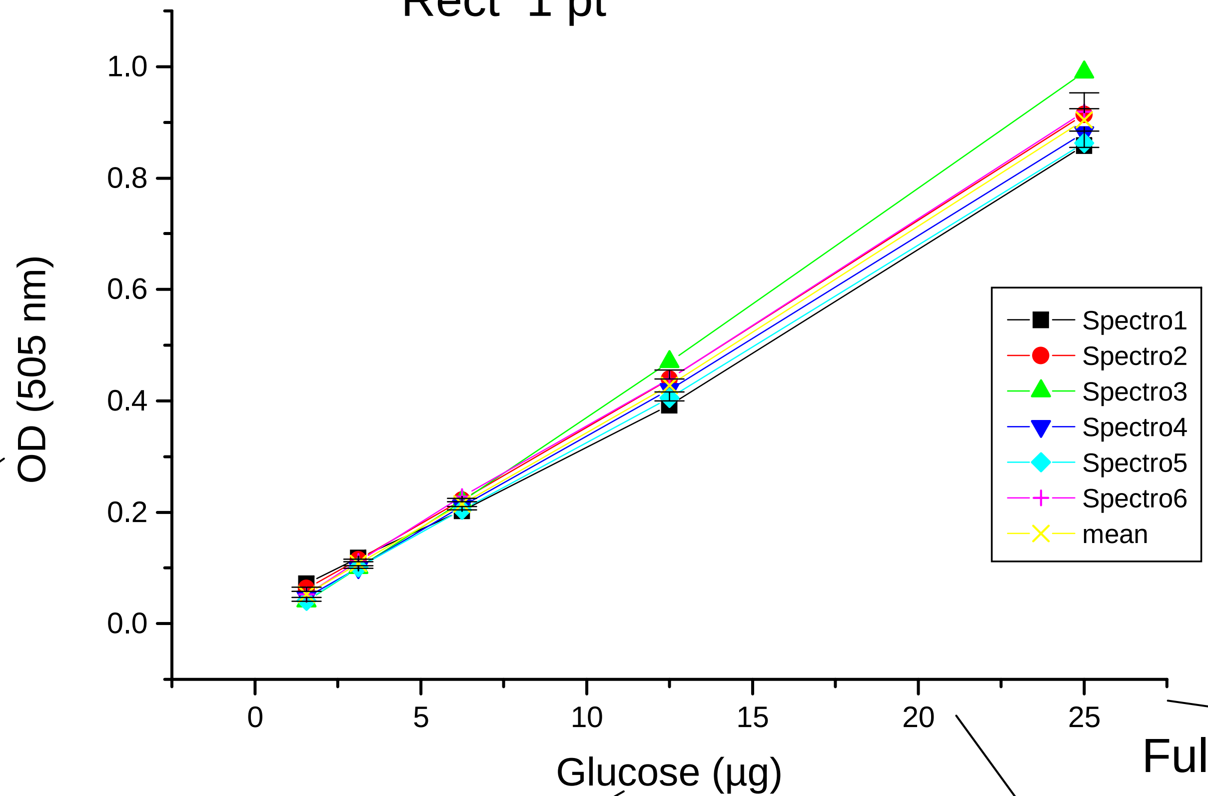
Total 10 points

Values properly plotted 5 pt

Rect 1 pt

Units 0,5 pt

Title 0,5 pt



Full range scale 1 pt

Linneal scale 1 pt

Title 0,5 pt

Units 0,5 pt

Total 10 points

5 points if interpolation is correct

Absorbance of the sample	X
Concentration of the sample (in µg . ml $^{-1}$)	1100-1400

5 points

Question 2 (3 points)

A 3 points

Question 3 (1 point)

X AB 1 point XB 0.5 point XA 0.5 point

Question 4 (9.6 points)

A	XB	XC	XD
0.9	0.9	0.9	0.9

XA	B	XC
2	2	2

Question 5 (4 points)

XC

17^{th} INTERNATIONAL BIOLOGY OLYMPIAD
9 - 16 JULY 2006
Río Cuarto – República Argentina

PRACTICAL TEST

4

MICROBIOLOGY

Student Code:	

17 th INTERNATIONAL BIOLOGY OLYMPIAD
9-16 JULY 2006
Río Cuarto – República Argentina

General remarks about the practical tests

DEAR PARTICIPANTS

The practical tests are organized in four different laboratories.

Nº 1- Plant Anatomy, Systematics and Physiology

Nº 2- Animal Anatomy, Ecology and Systematics

Nº 3- Biocheminstry

Nº 4- Microbiology

- You have **1 hour** for each laboratory: Nº 1 and Nº 2.
- You have **1 hour 30 minutes** for each laboratory: Nº 3 and Nº 4.
- You can score maximum of **40 points** in each laboratory, which means a total of **160 points** for the whole practical test.

Good luck !!!!!!!

Practical test 4

MICROBIOLOGY

There are different systems of bacterium classification, but the most commonly used is the published in Bergey's Manual of Determinative Bacteriology.

A working outline for the identification of a bacterial strain from the biochemical point of view is proposed below:

1) Isolate the strain and obtain a pure culture.

2) Carry out a microscopic examination of living cells and also of Gram stained smears. The morphology and type of Gram staining of the microorganism under study is thus determined. It is also important to determine the presence of clusters, spores and any other morphological characteristics that may be of interest.

3) Determine the nutritional characteristics (in general they come off from the methods used in the previous isolate and culture): photoautotrophs, photoheterotrophs, chemiautotrophs, chemiheterotrophs.

4) Conduce primary tests: The following group of tests, called primary tests, are used to determine the genus, group of genera or in some cases, the family to which an isolate belongs to. The primary tests are, beside Gram staining and morphology observation, the determination of catalase, oxidase, glucose fermentation, and motility, among others.

Reagents and Equipment

1. Dropping bottle with Gentian Violet (ready to use)
2. Dropping bottle with Lugol (ready to use)
3. Dropping bottle with Gram decolorizer (ready to use)
4. Dropping bottle with Safranin (ready to use)
5. Dropping bottle with Distilled water
6. 1 tube rack
7. 2 Kahn tubes containing a culture grown in Luria-Bertani medium of organisms A and B.

8. 2 Lab gloves
9. Respiratory mask
10. Marker pen
11. Paper napkin
12. 1 Bunsen burner
13. Microscope
14. Loop
15. 4 Slides
16. Tray with slide holder
17. 1 plastic bottle with water for rinsing
18. 1 disposable glass
19. 1 dropping bottle with immersion oil
20. 1 dropping bottle with 3% H_2O_2
21. 2 Luria-Bertani agar plates inoculated with organisms A and B.
22. 1 Eppendorf tube with 2 oxidase disks
23. 1 pair of tweezers
24. 2 Kahn tubes
25. 1 Kahn tube with a stopper containing sterile distilled water.
26. 1 plastic Pasteur pipette.
27. 3 plates with eosin methylene blue agar (EMB) medium (one of them inoculated with organism A, another inoculated with organism B and the last one without inoculation)
28. 3 tubes with phenylalanine (one of them inoculated with organism A, another inoculated with organism B and the last one without inoculation).
29. 1 Dropper containing 10% ferric chloride
30. 3 Kahn tubes with SIM (hydrogen Sulfide Indole Motility) medium (one of them inoculated with organism A, another inoculated with organism B and the last one without inoculation)
31. 1 Dropping bottle containing Indole reagent
32. 3 Kahn tubes containing UREA broth (one of them inoculated with organism A, another inoculated with organism B and the last one without inoculation)

33. 3 Kahn tubes with motility indole ornitine medium (MIO) (one of them inoculated with organism A, another inoculated with organism B and the last one without inoculation)
34. 3 Kahn tubes containing Simmons citrate (labeled as SC-A and SC-B and another one labeled SC without inoculation)
35. Clock located in view of all the students in the laboratory.

Caution:

You must be careful in the manipulation of media and reagents since the quantities provided allow performing this practical test only once.

If you work carelessly, with abrupt movements, far from the burner, you will contaminate the medium thereby preventing you from obtaining good results.

You will perform the biochemical tests which basis and interpretation are detailed below by using the media, reagents, and the given bacteriological information (charts and diagrams)

Note: Do not discard the tubes with organisms A and B. You will use them in task 2.

TASK 1: Perform Gram-staining in organisms A and B.

EXPERIMENTAL PROCEDURE

Introduction:

Gram stain differentiates between two major bacterial cell wall types. Some bacterial species, because of the chemical nature of their cell walls, have the ability to retain the crystal violet even after the treatment with an organic decolorizer such as a mixture of acetone and alcohol.

Gram stain technique

1. Make a thin smear of the material to study and allow to air dry.
2. Fix the material to the slide so that it does not wash off during the staining procedure by passing the slide three or four times through the flame of a Bunsen burner.
3. Place the smear on a staining rack and overlay the surface with Gentian Violet solution.
4. After 30 seconds of exposure to the Gentian Violet solution, wash thoroughly with running water.
5. Next, overlay the smear with Gram's iodine solution (lugol) for 30 seconds.
6. Hold the smear between the thumb and forefinger and flood the surface with a few drops of the acetone alcohol decolorizer until no violet color washes off. This usually requires 10 seconds or less time.
7. Wash with running water and again place the smear on the staining rack. Overlay the surface with safranin counterstain for 20 seconds. Wash with running water.
8. Place the smear in an upright position in a staining rack, allowing the excess water to drain off and the smear to dry.
9. Examine the stained smear under the 100 x (oil) immersion objective of the microscope.
10. When you focus the microscope call the assistant.

Results

SELECT THE CORRECT ANSWER, FILLING THE CORRESPONDING BOX

Organism	Gram staining		Assistant revision
A	☐ Positive	☐ Negative	
B	☐ Positive	☐ Negative	

~~**Question**~~

~~**Gram-variability**~~

~~**A)** is a term which can be used where two Gram reactions are seen due to an error in the staining procedure.~~

~~**B)** applies to an organism which changes its cell wall structure from the Gram-positive type to the Gram-negative type as the culture ages.~~

~~**C)** applies to what is ultimately seen when cells in a culture of gram-positive bacteria lose the ability to retain the primary stain during the decolorization process.~~

~~**D)** indicates a mixed (i.e., impure) culture.~~

~~**Write the letter corresponding to the correct answer on the dotted line below:**~~

~~.......................................~~

TASK 2: Biochemical characterization of organisms A and B.

In this part of the practical work you will determine, by means of metabolic tests (already provided or performed by you), the family and genus of the two organisms labeled as A and B.

Catalase Reaction:

Some bacteria contain flavoproteins that reduce the oxygen resulting in the production of hydrogen peroxide (H_2O_2) or superoxide (O_2 -), which are extremely toxic since they are powerful oxidizers able to destroy the cellular constituents in a short time. Many bacteria possess enzymes that offer protection against these toxic compounds.

Technique

Perform the catalase test to organisms A and B by adding two drops of H_2O_2 to a bacterial suspension (3 loopfuls of the liquid culture labeled as LB-A and LB-B) placed on the slides.

Note: Write the obtained results in the Biochemical test table using + (for a positive reaction) or – (for a negative reaction)

Question

Which of the following reactions is carried out by the catalase enzyme?

1) $H_2O_2 + NADH + H^+ \rightarrow 2\ H_2O + NAD^+$
2) $H_2O_2 + H_2O_2 \rightarrow 2H_2O + O_2$
3) $O_2^- + O_2^- + 2H+ \rightarrow H_2O_2 + O_2$
4) $4\ O_2^- + 4H^+ \rightarrow 2H_2O + 3\ O_2$

Write the number corresponding to the correct answer on the dotted line below:

……………………………..

Oxidase Reaction:

Test used for the detection of the cytochrome-c-oxidase enzyme, which is present in different genera, e.g. *Pseudomonas* spp., *Neisseria* spp., *Moraxella* spp., *Vibrio* spp., *Aeromonas* spp.

Oxidase discs contain dimethyl-para-phenylene-diamine, which is the substrate of cytochrome-c-oxidase enzyme.

Organisms possessing this enzyme, in the presence of atmospheric oxygen and the substrate contained in the oxidase discs give a red-fuchsia color.

Technique

Perform the oxidase test to organism A and B according to the following instructions:

Oxidase test will be carried out using tubes. From a pure culture, prepare a heavy suspension in 0.2 ml of sterile distilled water, and add one oxidase disc.

Note: Prepare the bacterial suspension starting from 3 colonies of each one of the plates labeled as LB-A and LB-B respectively.

Results

Generally, within the first minute and at room temperature, positive results are detected. A delayed reaction, evidenced after 2 minutes must be considered a negative result.

Positive: discs show a red-fuchsia color.

Negative: no changes in disc color.

Note: Write the obtained results in the Biochemical test table using + (for a positive reaction) or – (for a negative reaction)

EOSIN METHYLENE BLUE (EMB) AGAR

This medium, is used for the selective isolation of fast growing and with scarcely nutritional requirements Gram-negative bacteria.

It allows the growth of all Enterobacteriaceae members.

Purpose

This medium combines the Holt-Harris formulation with the Levine´s one to improve the selective isolation of Enterobacteriaceae and other Gram-negative bacterium species. The differentiation between lactose and/or sucrose fermenter organisms from those organisms which do not ferment them is possible due to the presence of the indicators eosin and methylene blue. Also, these indicators inhibit the growth of several Gram-positive bacteria.

Many strains of *Escherichia coli* and *Citrobacter spp.* show colonies with a greenish metallic sheen in this medium.

Lactose and/or sucrose fermenter organisms show colonies with a dark center surrounded by a blue or pink color, while lactose and/or sucrose non fermenter organisms show colorless colonies.

This medium also allows the growth of different organisms in addition to the growth of the Enterobacteriaceae members, and may be generally differentiated by the appearance of their colonies.

Instructions

Using the EMB plates provided (labeled as EMB-A and EMB-B for organisms A and B respectively), determine the sucrose and/or lactose utilization for organisms A and B.

Note: Write the obtained results in the Biochemical test table using + (for a positive reaction) or – (for a negative reaction).

Question

Fermentation

A) results in production of acid and possibly gas from the breakdown of sugars.

B) is associated with the type of growth of facultative anaerobes in Thioglycollate Medium where growth is less dense in the anaerobic region.

C) is generally associated with a positive catalase reaction for an organism.

Write the letter/s corresponding to the correct/s answer/s on the dotted line below:

…………………………………

Phenylalanine agar (Tubes labeled as Ph)

Phenylalanine agar is recommended for the detection of production of phenylpyruvic acid from phenylalanine by deamination. A positive reaction results in a green coloration after the application of 10% ferric chloride.

Instructions

Add 4 or 5 drops of the ferric chloride solution to the phenylalanine slants agar tubes (labeled as Ph-A and Ph-B for organisms A and B respectively). As the reagent is added rotate the tubes. An intense green color appearing within 10 minutes indicates the presence of phenylpyruvic acid.

Note: Write the obtained results in the Biochemical test table using + (for a positive reaction) or – (for a negative reaction).

Hydrogen Sulfide indole motility medium (SIM) (Tubes labeled as SIM A and SIM B)

This medium is used for the detection of hydrogen sulfide, indole production, and motility in the same tube. Hydrogen sulfide production in this medium is originated from thiosulphate or sulphate reductases and not by cysteine desulfhidrases. Any blackening along the line of inoculation is considered as a positive hydrogen sulfide reaction, and it usually appears between 18-24 hours of inoculation. Motile cultures in SIM medium show diffuse growth away from the line of inoculation. This is an appropriated medium for the detection of Listeria's characteristic "umbrella-like" movement. The high content of tryptophan in this medium makes it very suitable for detection of indole production.

Instructions

Using the SIM tubes provided (labeled as SIM-A and SIM-B for organisms A and B respectively), determine the production of hydrogen sulfide and indole, as well as the motility for organisms A and B.

For indole production detection, add 5 drops of the reagent (labeled as indole) to the heavy growth obtained in SIM tubes. A pink color promptly developed indicates the presence of indole.

Note: Write the obtained results in the Biochemical test table using + (for a positive reaction) or – (for a negative reaction).

Question

A negative result for motility

A) is indicated if growth occurs only along the line where the medium was stab-inoculated.

B) should be confirmed by a wet mount of a young culture of the same organism.

C) may exhibit growth over the surface of the medium.

D) may occur for strictly aerobic, motile organisms.

Write the letter/s corresponding to the correct/s answer/s on the dotted line below:

………………………………

UREA BROTH

This medium is suitable for the differentiation of urease producing organism.

Instructions

Using the urea broth tubes provided (labeled as UREA-A and UREA-B for organisms A and B respectively), determine the production of urease for organisms A and B.

Note: Write the obtained results in the Biochemical test table using + (for a positive reaction) or – (for a negative reaction).

SIMMONS CITRATE AGAR

It is a medium capable to differentiate between bacteria harboring citrate permease enzymes and those that do not harbor such enzymes.

Instructions

Using the SIMMONS CITRATE tubes provided (labeled as SC-A and SC-B for organisms A and B respectively), determine the presence of growth for organisms A and B.

Note: Write the obtained results in the Biochemical test table using + (presence of growth) or – (absence of growth).

Motility Indole Ornitine (MIO) Medium

The reactions in this medium are observed as follows:

- Ornithine Decarboxylation (ODC). Observe the lower three-quarters (anaerobic region) of the medium for change in color of the pH indicator; growth must be present in this part of the tube for correct analysis of result:
    - Gray, blue or purple color: Positive reaction for ornithine decarboxylation – formation of a highly alkaline product, over-neutralizing the acid produced from glucose fermentation.
    - Yellow color: Negative reaction. Yellow color is due to the "default" acid production from glucose fermentation.

Instructions

Using the MIO tubes provided (labeled as MIO-A and MIO-B for organisms A and B respectively), determine the production of ornithine decarboxylase enzyme for organisms A and B.

Note: Write the obtained results in the Biochemical test table using + (for a positive reaction) or – (for a negative reaction).

Results:

Write the results of the biochemical tests in the following table (**11 points**)

Organism	catalase	lactose	sucrose	motility	indole	H_2S	Phenyl alanine	ODC	Ureasa	citrate	oxidase
A											
B											

Using the tables in the annex indicate (**9 points**)

	Family	Genus
Organism A		
Organism B		

Questions

1. You have cultures of five organisms as listed below. However, the labels of the tubes have come off and you need to re-label the tubes correctly! First, you must consider the various reactions you know for the organisms in question:

genus	Gram stain	shape	catalase reaction	glucose fermentation	lactose fermentation	phenylalanine deaminase	citrate utilization
Bacillus	+	rod	+	+ or –	?	?	?
Staphylococcus	+	coccus	+	+	?	?	?
Enterobacter	–	rod	+	+	+	–	+
Morganella	–	rod	+	+	–	+	–
Pseudomonas	–	rod	+	–	–	–	?

a. The results obtained from what specific laboratory procedure will differentiate *Bacillus* and *Staphylococcus* from each other and also from the remaining three genera?

A) Glucose fermentation

B) Citrate utilization

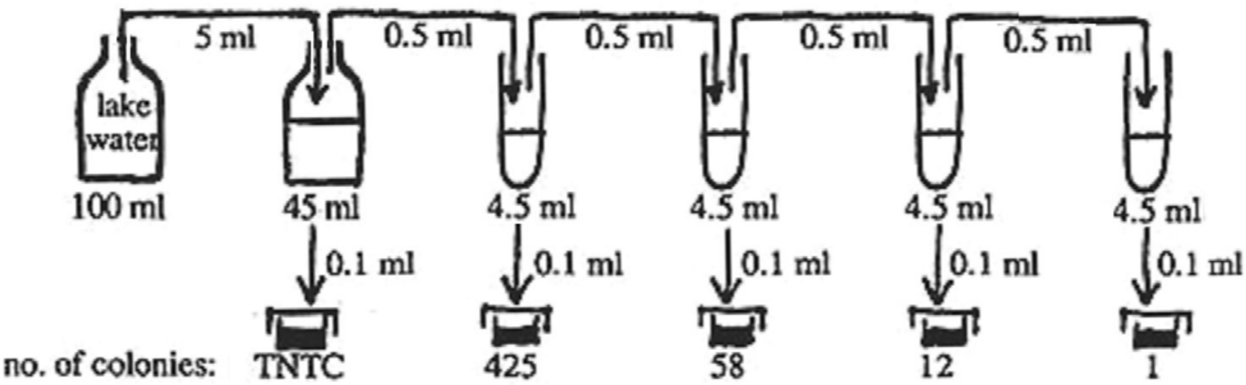
C) Catalase reaction

D) Gram stain

Write the letter corresponding to the correct answer on the dotted line below:

……………………………….

2. Consider the following dilution scheme:



a. Report the total number of CFUs (colony forming units) in the entire 100 ml amount of the original lake water sample. (TNTC=too numerous to count.)

A) 5.8 10^{7} cfu / 100 ml

B) 4.25 10^{8} cfu / 100 ml

C) 1.2 10^{9} cfu / 100 ml

Write the letter corresponding to the correct answer on the dotted line below:

…………………………………

b. Would you expect any change in the answer of the above problem if the first dilution was made by adding one ml of sample to 9 ml of diluent?

A) Yes

B) No

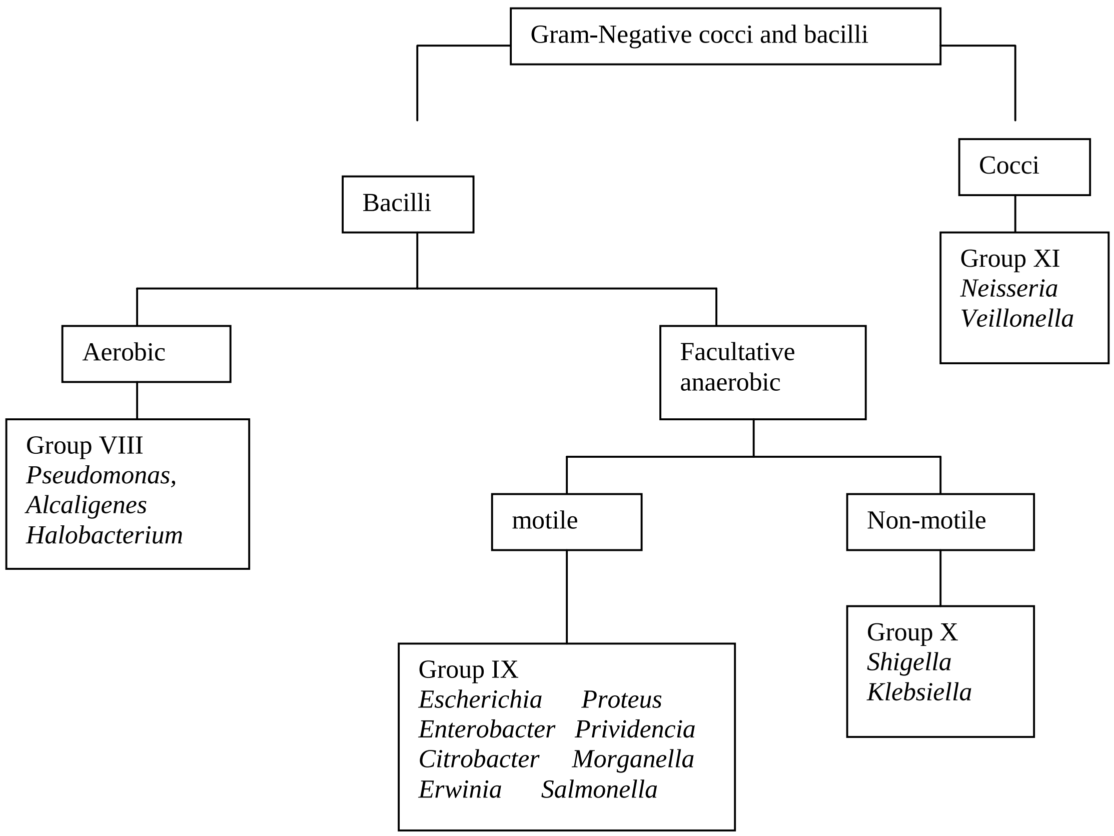
Write the letter corresponding to the correct answer on the dotted line below:

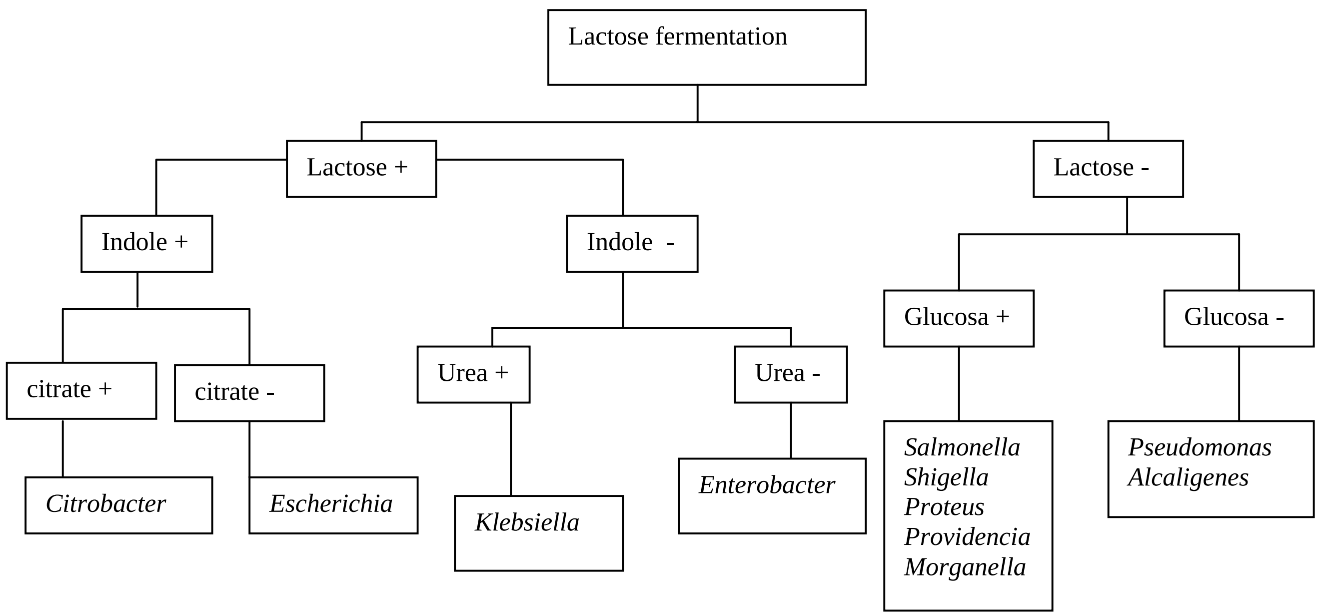
…………………………………

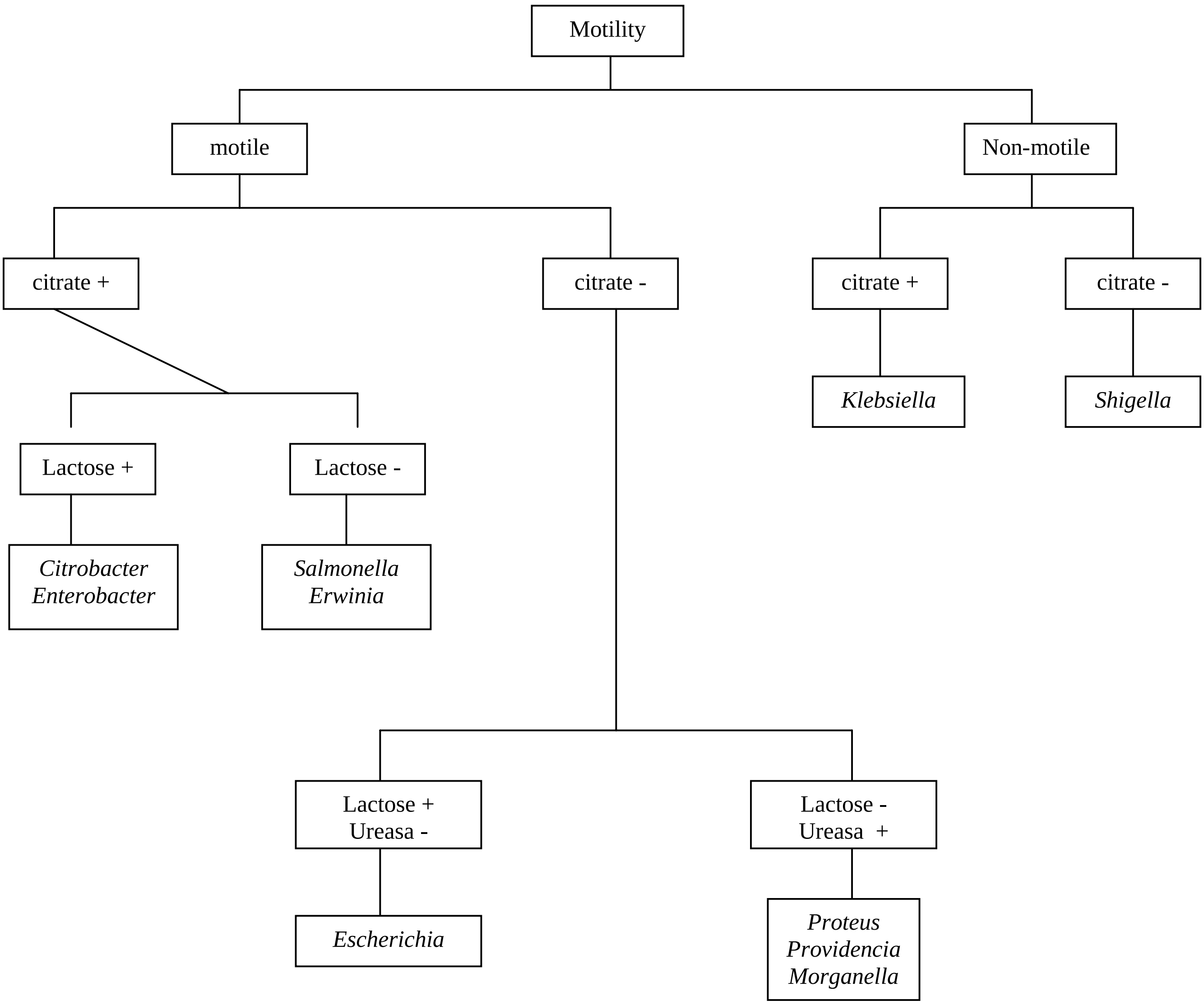
Annex 1

Gram stain (fresh culture)	+	+	+	+	+	+	+	+	–	–	–	–	–
shape	coccus	coccus	coccus	coccus	rod	rod	rod	rod	rod	rod	rod	rod	coccus
grouping	clusters	clusters	chains	Tetrads									pairs
aerobic growing	+	+	+	+	+	–	+	+	+	+	+	+	+
anaerobic growing	–	+	+	+	+	+	+	–	–	–	+	+	–
motility	–	–	–	–	–	+ or –	+ or –	+ or –	+ or–	+ or –	+ or -	+	–
catalase	+	+	–	–	–	–	+	+	+	+	+	+	+
oxidase									+	+	–	+	+
fermentation of glucose to acid or acid+gas	–	+	+	+	+	+ (or –)	+	–	–	–	+	+	–
Micrococcus	**X**												
Staphylococcus		**X**											
Streptococcus			**X**										

Lactococcus			X										
Enterococcus			X										
Clostridium						X							
Bacillus							X	X					
Alcaligenes									X				
Pseudomonas										X			
Enterobacterias											X		
Aeromonas												X	
Chromobacterium												X	
Neisseria													X


Gram-Negative cocci and bacilli
Bacilli
Cocci
Group XI
Neisseria
Veillonella
Aerobic
Facultative anaerobic
Group VIII
Pseudomonas,
Alcaligenes
Halobacterium
motile
Non-motile
Group IX
Escherichia *Proteus*
Enterobacter *Prividencia*
Citrobacter *Morganella*
Erwinia *Salmonella*
Group X
Shigella
Klebsiella


Lactose fermentation
Lactose +
Lactose -
Indole +
Indole -
Glucosa +
Glucosa -
citrate +
citrate -
Urea +
Urea -
Citrobacter
Escherichia
Klebsiella
Enterobacter
Salmonella
Shigella
Proteus
Providencia
Morganella
Pseudomonas
Alcaligenes


Motility
motile
Non-motile
citrate +
citrate -
citrate +
citrate -
Klebsiella
Shigella
Lactose +
Lactose -
Citrobacter
Enterobacter
Salmonella
Erwinia
Lactose +
Ureasa -
Lactose -
Ureasa +
Escherichia
Proteus
Providencia
Morganella

Family	Genus		oxidation									
		catalase	lactose	sucrose	motility	indole	SH_2	Phenyl alanine	ODC	Ureasa	citrate	oxidase
Enterobacteriaceae	*Escherichia*	+	+	+	+	+	-	-	-	-	-	-
	Shigella	+	-	-	-	+	-	-	-	-	-	-
	Salmonella	+	-	-	+	-	+	-	+	-	-	-
	Citrobacter	+	+ / -	-	+	+	+	-	+	-	+	-
	Proteus	+	-	+	+	+ / -	+	+	+	+	+ / -	-
	Morganella	+	-	-	+	+	-	+	+	+	-	-
	Enterobacter	+	+	+ / -	+	-	-	-	+	+ / -	+	-
	Serratia	+	+ / -	+ / -	+	-	-	-	+ / -	+	+	-
	Klebsiella	+	+	+	-	-	-	-	-	+	+ / -	-
Pseudomonaceae	*Pseudomonas*	+	?	-	+	-	-	+ / -	+ / -	+ / -	+ / -	+

Correct Answer Sheet

Practical test 4

MICROBIOLOGY

TASK 1: Perform Gram-staining in organisms A and B.

Organism	Gram staining		Assistant revision
A	☐ Positive	**X** Negative	
B	☐ Positive	**X** Negative	

10 point (5 point organism A and 5 point organism B)

Question

TASK 2: Biochemical characterization of organisms A and B.

Question (1.6 point)

Which of the following reactions is carried out by the catalase enzyme?

………………2…………………

Question

Fermentation **(1.6 point)**

…………………A, B………………

Question

A negative result for motility **(1.6 point)**

………A, B, C, D…………………………

Results:

Write the results of the biochemical tests in the following table (11 point)

Organism	catalase	lactose	sucrose	motility	indole	SH_2	Phenylalanine	ODC	Ureasa	citrate	oxidase
A	**+ (0.5p)**	**+ (0.5p)**	**+ (0.5p)**	**+ (0.5p)**	**+ (0.5p)**	**- (0.5p)**	**- (0.5p)**	**- (0.5p)**	**-(0.5p)**	**- (0.5p)**	**- (0.5p)**
B	**+ (0.5p)**	**- (0.5p)**	**- (0.5p)**	**+ (0.5p)**	**+ (0.5p)**	**- (0.5p)**	**- / + (0.5p)**	**+ / - (0.5p)**	**-(0.5p)**	**+ (0.5p)**	**+ (0.5p)**

Using the tables in the annex indicate (9 point)

	Family	Genus
Organism A	***Escherichiae* (2.25 p)**	***Escherichia* (2.25p)**
Organism B	***Pseudomonaceae* (2.25p)**	***Pseudomonas* (2.25p)**

Questions

1. (**1.6 point)**

………………D…………………

2. a. **(1.6 point)**

………………A…………………

b. (2 point)

………………B…………………